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In [3]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

FILE = r"D:\Datascience-MachineLearning\Datasets\diabetes.csv"

df = pd.read_csv(FILE)
X = df.iloc[:, :-1]          # features
y = df.iloc[:, -1]          # last column = Outcome (0 or 1)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
cm = confusion_matrix(y_test, pred)
print("Confusion Matrix:")
print(cm)

# Confusion matrix as an image
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["No Diabetes (0)", "Diabetes (1)"])
disp.plot(cmap="Blues", values_format="d")
plt.title("Confusion Matrix - Logistic Regression (Diabetes)")
plt.savefig("confusion_matrix.png", dpi=150, bbox_inches="tight") # saves as an image file
plt.show()

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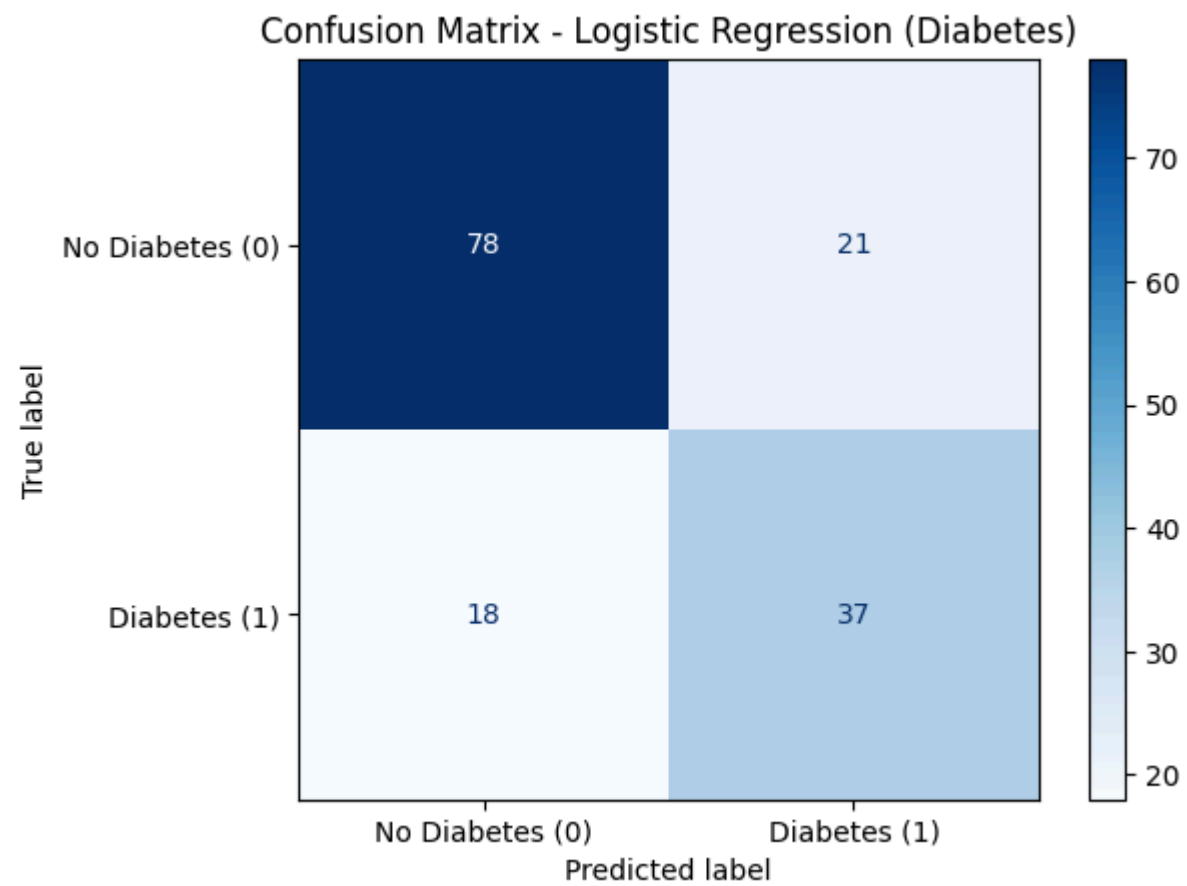
Accuracy: 0.7467532467532467

Confusion Matrix:

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[[78 21]
 [18 37]]

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In []: